

First Terminal Examination - 2082

Class : XI	Subject: Biology (Code: 2011)	Full Marks: 75
Time: 3 hrs.	SET 'A'	Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

1. Which one of the following is wrong?
 - a. Taxonomy is the functional science which deals with identification, nomenclature and systematic classification of organisms
 - b. Scientific name is the name given to an organism by the scientists on the basis of certain criteria and rules so that it can be identified by same name throughout the world
 - c. Scientific names are necessary whenever precise identification is required
 - d. In botany, tautonyms are allowed.
2. The ecology of individual species is
 - a. Synecology
 - b. Autecology
 - c. Ecosystem ecology
 - d. Human ecology
3. Which one of the following is not edaphic factor?
 - a. Steepness of slope
 - b. Soil water
 - c. Soil organic matter
 - d. Soil texture & soil particle
4. Biological Oxygen Demand (BOD) is
 - a. Oxygen required by living organisms for respiration in an ecosystem
 - b. Oxygen required by aquatic plants for their respiration
 - c. Oxygen required by aquatic animals for their respiration
 - d. Oxygen required by decomposers to decompose the dead bodies in an ecosystem

5. Which one of the following is also called osmotrophs ?
 - a. Decomposers
 - b. Producers
 - c. Primary consumers.
 - d. Tertiary consumers
6. The structure of an ecosystem is related to
 - a. Operation of ecosystem
 - b. Species diversity
 - c. Energy flow
 - d. Mineral cycling

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

7. Define binomial system. Write down the rules of binomial system of nomenclature. [1+3]
8. Define ecology. Describe the ecological role of light on flora and fauna. [1+3]
9. Describe mutualism and social organization with examples. [2+2]
10. Define cell biology. Write down the postulation of cell theory. [1+3]

OR

Give appropriate scientific term for each of the following.

 $[1 \times 4]$

- The arrangement of taxonomic categories in their proper order on the basis of their taxonomic ranks
- The interaction between the organisms of two different species where both the interacting partners are benefited & the relation is non – obligatory
- The ability of a living, somatic, nucleated cell to form the complete organism
- The transfer of food energy from producers to the different grade of consumers animal with repeated process of eating & being eaten

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

11. Describe the structural & functional aspects of pond ecosystem. [4+4]
12. What is the function of Pyrenoid in *Spirogyra*? Describe the life cycle of *Spirogyra*. [1+7]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

13. Term biology was introduced by
- | | |
|---------------------------|--------------|
| a. Lamarck and Treviranus | b. Darwin |
| c. John Ray | d. Aristotle |
14. Charles Darwin's theory of evolution is based on:
- | | |
|----------------|-----------------------|
| a. Environment | b. Chromosome pairing |
| c. Mutation | d. Natural Selection |
15. Primordial soup is a set of hypothetical conditions on ancient earth first proposed by
- | | |
|----------------------|---------------------|
| a. Dmitri Ivanosvsky | b. Alexander Oparin |
| c. Dmitry Anuchin | d. Nikolay Shatsky |
16. The experiment that simulated conditions thought to be present on the early earth
- | | |
|-----------------------------|------------------------------|
| a. Hershey-Chase Experiment | b. Geiger-Marsden Experiment |
| c. Miller-Urey Experiment | d. Schiehallion Experiment |
17. Which of the following is connecting link between animals and plant kingdom?
- | | |
|--------------------|------------------|
| a. <i>Euglena</i> | b. <i>Amoeba</i> |
| c. Dinoflagellates | d. Diatoms |

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 =16]

18. Lamarckism is considered as one of the first theory in evolutionary biology. Criticize this theory as you have studied in this theory. [2+2]
19. *Archaeopteryx lithographica* is missing link between Reptiles and Aves, discuss. [2+2]
20. Summarize Miller's simulation experiment for organic synthesis with figure. Comment on its efficacy. [2+2]
21. How vestigial organ support the organic evolution? Describe it with figures. [3+1]

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

22. Which is the most accepted theory in origin of life? Briefly relate chemogeny, biogeny and cognogeny on origin of life. [1+3+2+2]
23. Name the theory of evolution purposed by Darwin. Enumerate the key concepts in the evolution theory of Darwin. [1+7]

❧ The End ❧